

## Project: Customized Routing simulator

Prerequisite: Python and object-oriented programming skills

SimPy is process-based discrete event simulation framework for Python [1]. It has the potential of simulating protocol dynamics of IoT networks.

This project designs and demonstrates a routing simulation using PySim. The aim is to evaluate the Routing over Low Power Lossy Networks (RPL) protocol [ 2 ]. The project may start with a simplified version of the RPL demonstrating the establishment of the DODAG i.e, the DIS and the DIO message (Fig. 3). Hereafter, the progress with the distribution of network prefixes using DAO messages and extending the DIO messages with the trickle algorithm [3].

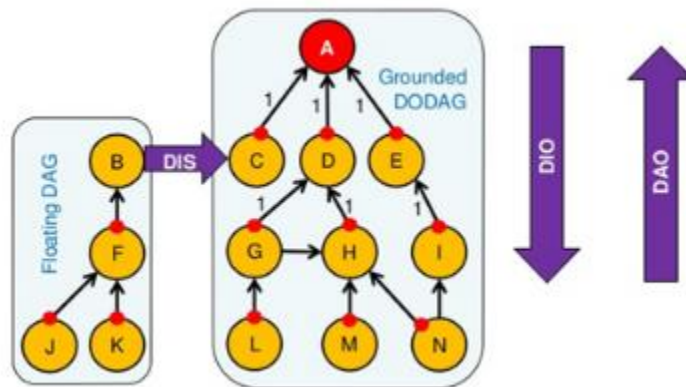


Figure 3: Example of the RPL routing protocol and its use of DIS and DIO messages to build and maintain the DODAG.

### Pivot points of the project

Try to address as many of the following points of the project.

- Establish a network and a node concept in SimPy.
- Implement and simulate a neighbor discovery mechanism that ensures that each node establish connectivity with its nearest neighbors.
- Implement the DIO message and simulate the DODAG formation with a varying number of nodes with different sets of neighbor nodes.
- Implement the DIS message (optional).
- Implement a trickle algorithm [3] to control the period of DIS messages in the network (optional).
- Design a node addressing plan for the network. Make assumption on the use of a mesh under or a route over architecture.
- Implement the DAO message and demonstrate the distribution of network prefixes in the DODAG.

- Simulate a DODAG local repair mechanism (optional).
- Organize and document your code for the simulator in a software repository such as Git.

## Reporting

The report should be written in a self-contained style addressing the general criteria listed in the introduction section. In your reporting, your reporting should address the following:

- A brief introduction to RPL.
- Clarify of the performance metrics used for the protocol evaluation.
- A description of the implementation.
- Simulation results and their analysis interpretation.

## References

- [1] SimPy, Discrete event simulation for Python, <https://simpy.readthedocs.io/>, accessed 2021-08-12.
- [2] T. Winter, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, JP. Vasseur, RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks, Internet Society, RFC 6550, March 2012.
- [3] P. Levis, T. Clausen, J. Hui, O. Gnawali, and J. Ko, The Trickle Algorithm, Internet Society, RFC 6206, March 2011.